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Molecular Opacity Calculations for Lithium Hydride at Low Temperature

Gui-Ying Liang(梁桂颖)[1], Yi-Geng Peng(彭裔耕)[2], Rui Li(李瑞)[3], Yong Wu(吴勇)[1,4*], and Jian-Guo Wang(王建国)[1]

[1]*Institute of Applied Physics and Computational Mathematics, Beijing 100088, China*
[2]*Department of Applied Physics, Nanjing University of Science and Technology, Nanjing 210094, China*
[3]*Department of Physics, College of Science, Qiqihar University, Qiqihar 161006, China*
[4]*HEDPS, Center for Applied Physics and Technology, Peking University, Beijing 100084, China*

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The opacities of the lithium hydride molecule are calculated for temperatures of 300 K, 1000 K, 1500 K, and 2000 K, at a pressure of 10 atm, in which the contributions from the five low-lying electronic states are considered. The *ab initio* multi-reference single and double excitation configuration interaction (MRDCI) method is applied to compute the potential energy curves (PECs) of the ^{7}LiH, including four $^1\Sigma^+$ states and one $^1\Pi$ state, as well as the corresponding transition dipole moments between these states. The ro-vibrational energy levels are calculated based on the PECs obtained, together with the spectroscopic constants. In addition, the partition functions are also computed, and are provided at temperatures ranging from 10 K to 2000 K for ^{7}LiH, ^{7}LiD, ^{6}LiH, and ^{6}LiD.

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The light elements, hydrogen and lithium, which are significant in relation to both fundamental astrophysics and studies of the early Universe, have been observed in the galactic center, and their corresponding abundance values have been calculated.[1–5] These discoveries have proved helpful and meaningful in terms of investigations into stellar structure and galactic evolution, as well as providing crucial information regarding the evolution of the early Universe.[6–9] At the same time, the ^{7}LiH molecule, a possible coolant in the gravitational collapse of cosmic background radiation,[8] is primarily formed by the radiative association process of $^7\mathrm{Li} + \mathrm{H} \rightarrow {}^7\mathrm{LiH} + \nu$. The cooling process of the ^{7}LiH molecule is predominantly generated by ro-vibrational transitions between nuclear and electron motion. Thus, the spectrum of ^{7}LiH and its isotopic molecules have long attracted a great deal of attention in the context of both experimental and theoretical studies.

As the simplest metal hydrides, ^{7}LiH and ^{7}LiD molecules have been studied extensively in various fields, such as astrophysics, atmospheric science, molecular spectroscopy, and quantum chemistry. The spectra of $A^1\Sigma^+ \rightarrow X^1\Sigma^+$ for ^{7}LiH and ^{6}LiH, primarily located in the region of 300–410 nm, were first observed by Nakamura as early as 1930.[10] Subsequently, high-precision experiments were performed to detect the corresponding spectrum of $A^1\Sigma^+ \rightarrow X^1\Sigma^+$,[11,12] which covers $\nu'' = 0$–2 and $\nu' = 1$–15, respectively. Absorption spectra of ^{7}LiH and ^{7}LiD were also observed in the near-ultraviolet region of 290.8–293.0 nm. The observed break-off of the rotational structure for the $B^1\Pi \rightarrow X^1\Sigma^+$ band system has been interpreted as predissociation, due to rotation.[13] Experimentally, the band system of the $B^1\Pi \rightarrow X^1\Sigma^+$ was identified as occurring in the region 288–308 nm. The maxima of the absorption was separately located at about 1500 cm^{-1} and 1000 cm^{-1} for ^{7}LiH and ^{7}LiD, respectively. The millimeter spectrum of the rotational transition line of $J = 1 \leftarrow 0$ in the ground state and the first excited vibrational state for ^{7}LiD has been observed by Pearson and Gordy[14] and lines of $J = 1 \leftarrow 0$ with respect to ^{7}LiH, and $J = 2 \leftarrow 1$ for ^{7}LiD were measured in the submillimeter wave region by Plummer *et al.*[15,16] Since then, the rotational and vibrational spectra of ^{7}LiH and ^{7}LiD have been extensively studied by Maki and Thompson in the region of the far-infrared and mid-infrared.[17] In 1994, the spectroscopy of $J = 1$ to 10 ($\nu = 0$) was measured by Matsushima *et al.*[18] for ^{7}LiH, with the intention of filling the deficiency between previous submillimeter and far-infrared observations. The high-resolution Fourier transform spectra of ^{7}LiH and ^{7}LiD were detected in the far-infrared and mid-infrared regions by Dulick *et al.*,[19] while the first overtone ($\Delta\nu = +2$) rotational lines for the ground state of ^{7}LiH were measured successfully for the first time. Subsequently, the ro-vibrational levels of the $C^1\Sigma^+$ electronic state, including rotational structures and vibrational level spacings, were observed for the first time by Lin *et al.*[20] More recently, the spectra of the $D^1\Sigma^+$ state was observed by means of measured rotational structure and term values.[21] The $C^1\Sigma^+$ excited electronic state of ^{6}LiH and ^{7}LiD isotopomers has been studied,

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*Corresponding author. Email: wu_yong@iapcm.ac.cn
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using a pulsed optical-optical double resonance fluorescence depletion spectroscopic technique: forty-two vibrational levels ($\nu = 2$–43) and fifty-five vibrational levels ($\nu = 4$–58) were measured, respectively.[22] In conjunction with the previous observation of the ^{7}LiH, a combined isotopomer Dunham-type analysis was then performed.

On the theoretical side, earlier studies focused on ground state calculations.[23,24] Thereafter, low-lying excited state calculations, including potential energy curves (PECs), electronic wavefunctions, and dipole moments, were also performed.[25–28] The low vibrational levels of $X^1\Sigma^+$ and $A^1\Sigma^+$ states, together with radiative transition probabilities for the $B^1\Pi \rightarrow X^1\Sigma^+$ and $B^1\Pi \rightarrow A^1\Sigma^+$ emission bands of ^{7}LiH were also calculated.[29] Based on the existing spectral information, Vidal *et al.*[30] recalculated the PECs and molecular constants, taking account of the adiabatic corrections, and reanalyzed the spectra of the $A^1\Sigma^+ \rightarrow X^1\Sigma^+$ system of the isotopic lithium hydrides ^{6}LiH, ^{7}LiH, ^{6}LiD, and ^{7}LiD. Combining the pseudo-potential of core electrons and the full configuration interaction (CI) method for valence electrons, Boutalib *et al.*[31] computed the adiabatic and diabatic PECs of ^{7}LiH. Using the single-reference coupled-cluster method, Lee *et al.*[32] investigated its ground state PEC, as well as providing the corresponding spectral data. The dipole moments were computed to a high degree of accuracy by means of all-particle explicitly correlated Gaussian functions with shifted centers in Ref. [33], and the non-adiabatic effects on the spectroscopic properties of ^{7}LiH were also investigated. In 2011, Holka *et al.*[34] calculated the PECs and vibrational energy levels for the electronic ground state of ^{7}LiH, using Boron–Oppenheimer breakdown corrections. The ro-vibrational spectra of the ground state of ^{7}LiH were calculated by Coppola *et al.*[35]

Despite an extensive body of works relating to lithium hydride, previous measurements or calculations have tended to focus on the structure and spectra of the ground state.[35–37] As a result, studies relating to excited states, which are significant in terms of the high temperature conditions of fusion and astrophysical environments,[38–42] are few in number. In this Letter, we perform systematic calculations for ^{7}LiH in the five electronic states, including PECs, dipole moments, and transition dipole moments, as well as the ro-vibrational resolved opacities for ^{7}LiH, ^{7}LiD, ^{6}LiH, and ^{6}LiD, at temperatures in the range 300–2000 K.

Method. In this work, the multi-reference single and double excitation configuration interaction (MRDCI) package[43,44] is used to compute the PECs of the ground state, together with four low-lying singlet excited states: $X^1\Sigma^+$, $A^1\Sigma^+$, $B^1\Pi$, $C^1\Sigma^+$, and $D^1\Sigma^+$. Dunning's augmented correlation consistent polarized valence quadruple zeta, aug-cc-pVQZ[45] Gaussian basis sets are employed in relation to both lithium and hydrogen atoms. The correlation for the 1s of lithium is also considered in our calculation. Furthermore, the diffuse [3s3p3d] functions for Li and H are added in order to describe the excited states; the corresponding parameters are listed in Table S1 of the Supplementary Material. Overall, a [28s10p6d3f] contracted to [10s10p4d3f] basis set is generated for lithium, with a [10s6p5d] contracted to [7s6p5d] basis set generated for hydrogen. A threshold of 10^{-8} hartree is used to control the configurations.[46,47] An internuclear distance range of 2.00–45.00 a.u. is considered. The effects of spin-orbit coupling and the relativistic effect are ignored on the basis of their weak effect on lithium and hydrogen. The PECs are presented by linking the energy points of the electronic states, taking account of the non-crossing rule between two Λ–S states with the same symmetry.

Under the Born–Oppenheimer approximation condition, the electronic Schrödinger equation of diatomic molecules can be expressed as

$$\hat{H}_e \Psi_e(r, R) = E_e \Psi_e(r, R), \tag{1}$$

where $\hat{H}_e$ denotes the Hamiltonian of N electrons of diatomic molecules, $\Psi_e(r, R)$ and E_e represent the electronic eigenfunction and eigenvalue, respectively, and r and R are the coordinates of the electron and the nucleus, respectively. Thus,

$$\hat{\mathrm{H}}_e = \sum_{i=1}^{N} \left(-\frac{1}{2}\nabla_i^2\right) + \sum_{i=1}^{N} \left(-\frac{Z_a}{r_{ai}} - \frac{Z_b}{r_{bi}}\right) + \sum_{i<j=1}^{N} \left(\frac{1}{r_{ij}}\right) + \frac{Z_a Z_b}{R}, \tag{2}$$

where Z_a and Z_b represent the charge number of nucleus A and B, r_{ai} and r_{bi} denote the distances between the electron and the nucleus, and r_{ij} is the distance between the electron and electron.

The eigenvalues and wave functions of vibration-rotation states, which are calculated as far as the dissociation energies of the electronic states under consideration, can be obtained by solving the nuclear Schrödinger equation:

$$\left[-\frac{1}{2\mu}\frac{d^2}{dR^2} + E_e(R) + \frac{J(J+1)}{2\mu R^2}\right]\chi_{vJ}(R) = E_{vJ}\chi_{vJ}(R). \tag{3}$$

Here, $E_{vJ}(R)$ denotes the ro-vibrational energy level, $E_e(R)$ is the PEC of the electronic state; μ is the reduced mass, χ_{vJ} is the ro-vibrational wavefunction, and J is the rotational quantum number. The Einstein coefficient for radiation is determined by

$$A_{v'j',v''j''} = \frac{4}{3}\alpha^3 \omega^3_{v'j',v''j''} \frac{S_{J'J''}}{2J''+1}, \tag{4}$$

with

$$S_{J'J''} = |\langle \chi_{v'J'}|D(R)|\chi_{v''J''}\rangle|^2 \phi_{J'J''}, \tag{5}$$

where α is a fine-structure constant, $\omega_{v'j',v''j''}$ represents the transition frequency between different

vibration-rotation levels, $v'J'$ and $v''J''$ denote the vibrational and rotational quantum number of lower and upper states, $S_{J'J''}$ is the line strength, $D(R)$ is the transition dipole moment function, and $\phi_{J'J''}$ is the Hönl–London factor given by

$$\phi_{J',J''} = \begin{cases} \Delta\Lambda = 0, & \frac{(J''+\Lambda'')(J''-\Lambda'')}{J''}, \\ \Delta\Lambda = +1, & \frac{(J''-1-\Lambda'')(J''-\Lambda'')}{4J''}, \\ \Delta\Lambda = -1, & \frac{(J''-1+\Lambda'')(J''+\Lambda'')}{4J''}, \end{cases} \quad (P \text{ branch})$$

$$\phi_{J',J''} = \begin{cases} \Delta\Lambda = 0, & \frac{(2J''+1)^2\Lambda''^2}{J''(J''+1)}, \\ \Delta\Lambda = +1, & \frac{(J''+1+\Lambda'')(J''-\Lambda'')(2J''+1)}{4J''(J''+1)}, \\ \Delta\Lambda = -1, & \frac{(J''+1-\Lambda'')(J''+\Lambda'')(2J''+1)}{4J''(J''+1)}, \end{cases} \quad (Q \text{ branch}) \tag{6}$$

$$\phi_{J',J''} = \begin{cases} \Delta\Lambda = 0, & \frac{(J''+1+\Lambda'')(J''+1-\Lambda'')}{J''+1}, \\ \Delta\Lambda = +1, & \frac{(J''+2+\Lambda'')(J''+1+\Lambda'')}{4(J''+1)}, \\ \Delta\Lambda = -1, & \frac{(J''+2-\Lambda'')(J''+1-\Lambda'')}{4(J''+1)}, \end{cases} \quad (R \text{ branch})$$

where Λ is the component of the total angular momentum on the molecular axis, and J is the rotational quantum number.

Molecular opacity is calculated based on the integrated line strengths in Ref. [48], given by

$$\sigma = \frac{1}{8\pi(E''/hc)^2} \frac{A(2J'+1)\exp(-\Delta E_{V'J',00}hc/kT)[1-\exp(-E''/kT]}{Q}, \tag{7}$$

in which Einstein's coefficient A is obtained from Eq. (4), E'' denotes the energy gap of the two states concerned, h is Planck's constant, c is the speed of light in vacuum, k is the Boltzmann constant, and $\Delta E_{V'J',00}$ is the excitation energy of the lower state, measured in cm^{-1}. The total internal partition function $Q(T)$ is defined by summing up a specific electronic state, weighed by the Boltzmann factor, by means of

$$Q = \sum_{i=1}^{n} Q_{ei}Q_{vi}Q_{ri}\exp\left(-\frac{E_i}{kT}\right), \tag{8}$$

where T represents the environmental temperature, and Q_{ei} is a small parameter related to electron spin and orbital angular momentum; $Q_{vi} \times Q_{ri}$ is obtained by summing up all vibration-rotation states for the electronic states of i.

Results and Discussion—Potential Energy Curves and Spectroscopic Data. Here, we compute the first five low-lying bound states of the ^{7}LiH molecule, covering the four lowest dissociation limits, $^7\mathrm{Li}(1s^22s)+\mathrm{H}(1s)$, $^7\mathrm{Li}(1s^22p)+\mathrm{H}(1s)$, $^7\mathrm{Li}(1s^23s)+\mathrm{H}(1s)$, and $^7\mathrm{Li}(1s^23p)+\mathrm{H}(1s)$ at the asymptotic region. Only the singlet states are considered in this work, due to the selection rule. The PECs of the ^{7}LiH molecule are plotted in Fig. 1, in which the ground state $X^1\Sigma^+$ is associated with the asymptotic limit of $^7\mathrm{Li}(1s^22s)+\mathrm{H}(1s)$, while $A^1\Sigma^+$ and $B^1\Sigma^+$ states correspond to the asymptotic limit of $^7\mathrm{Li}(1s^22p)+\mathrm{H}(1s)$. The $C^1\Sigma^+$ state has two minima, around $R = 3.76$ a.u. and $R = 10.18$ a.u., resulting from the avoided crossing, while $D^1\Sigma^+$ dissociates into $^7\mathrm{Li}(1s^23p)+\mathrm{H}(1s)$ in the asymptotic region.

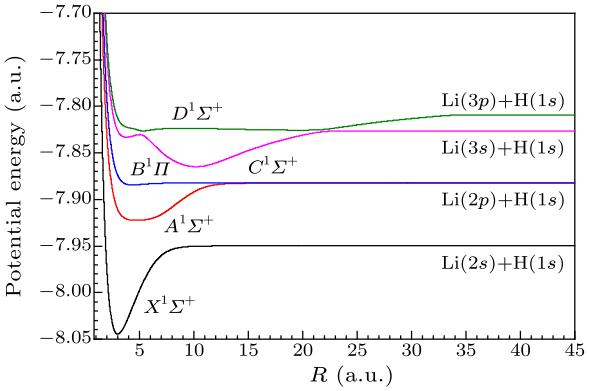


Fig. 1. Potential energy curves of five low-lying states of ^{7}LiH.

In order to check the accuracy of the PECs calculations, we evaluate the spectroscopic constants of the five electronic states under consideration, including transition energies T_e, rotational constant (B_e), vibrational frequencies (the harmonic frequency ω_e, and the anharmonic constant $\omega_e x_e$), equilibrium distance R_e, and dissociation energy D_e. These are provided in Table 1, together with the corresponding experimental values[46,47] and previous calculations.[31,34,49,50] The results in Table 1, indicate that our calculations are in good agreement with available data, both

experimental[46,47] and theoretical.[31,34,49–51] For the ground state $X^1\Sigma^+$, there is a potential well with a D_e value of 2.509 eV, which agrees well with the experimental data of 2.515 eV[48] and the theoretical result of 2.516 eV[34] obtained using the Born–Oppenheimer breakdown corrections. The B_e and R_e values of the ground state $X^1\Sigma^+$ are 7.0513 cm^{-1} and 3.021 Å, respectively, which deviate from the experimental data only by 0.4618 cm^{-1} and 0.006 Å.[47] The ω_e and $\omega_e x_e$ values obtained, being 1433.54 cm^{-1} and 24.01 cm^{-1}, are also consistent with the experimental results.[47] For the first excited state, $A^1\Sigma^+$, the dissociation energy value obtained in our calculations is 1.051 eV, which agrees well with the experimental value of 1.076 eV[47] and theoretical values of 1.048 eV[31] and 1.077 eV.[49] For the second excited state, $B^1\Pi$, our results for T_e, B_e, ω_e, $\omega_e x_e$ and R_e are all in good agreement with available experimental[47] and theoretical results.[49] However, for the excited state $C^1\Sigma^+$, the two potential wells are calculated at a depth of 0.145 eV and 1.077 eV, with the latter value deviating from the experimental data by about 0.027 eV, due to the avoided crossing with the $D^1\Sigma^+$ state. Moreover, as shown in Table 1, the results for T_e in the $A^1\Sigma^+$ and $B^1\Pi$ states, respectively, are 26610 cm^{-1} and 34996 cm^{-1}, which are in excellent agreement with the experimental measurements[47] of 26516 cm^{-1} and 34912 cm^{-1}, respectively. Based on the PECs obtained, the vibrational energy levels of the $X^1\Sigma^+$, $A^1\Sigma^+$, $B^1\Pi$, $C^1\Sigma^+$, and $D^1\Sigma^+$ states of ^{7}LiH are computed by solving the one-dimensional Schrödinger equation. The vibrational energy levels obtained in our work are listed in Tables S2–S6 of the Supplementary Material, demonstrating impressive consistency with the available experimental[13,21,22,46,52,53] and theoretical data.[25,31,34,49,50,54]

Table 1. Spectroscopic constants of bound states for ^{7}LiH.

State	Method	T_e (cm^{-1})	ω_e (cm^{-1})	$\omega_e x_e$ (cm^{-1})	B_e (cm^{-1})	R_e (Å)	D_e (eV)
$X^1\Sigma^+$	Present	0	1433.54	24.01	7.0513	3.021	2.509
	Expt.[a]	0	1405.65	23.20	7.5131	3.015	2.515
	TDCI.[b]	0	1387.4	22.24	7.35	3.049	2.411
	Full CI[c]	0				3.007	2.501
	Full CI[d]					3.003	2.523
	MRCI[f]	0				3.014	2.516
$A^1\Sigma^+$	Present	26610	283.40	9.01	2.4569	4.897	1.051
	Expt.[a]	26516	280.96	7.40	2.8536	4.906	1.076
	TDCI[b]		290.7	11.37	2.74	4.996	1.048
	Full CI[c]	26389.6				4.847	1.077
	Full CI[d]					4.862	1.077
$B^1\Pi$	Present	34996	126.0457	45.01	4.037	4.501	0.032
	Expt.[a]	34912	130.73	42.40	3.383	4.494	0.035
	TDCI[b]		171.13	54.10	3.11	4.688	0.017
$C^1\Sigma^+$	Present	46088				3.7	0.145
		38807				10.2	1.077
	Expt.[e]					10.14	1.050
	Full CI[c]	46109				3.825	0.158
		38942				10.21	1.047
	Full CI[d]					3.821	
						10.18	1.1
$D^1\Sigma^+$	Present	47287				5.25	0.341
		47419				19.56	
	Full CI[c]	47363				5.358	0.352
		47516				19.91	0.333
	Full CI[d]					5.28	0.464
						19.824	

[a]Ref. [47], [b]Ref. [49], [c]Ref. [31], [d]Ref. [50], [e]Ref. [46], [f]Ref. [34].

Electronic Transition and Permanent Dipole Moments. The transition and permanent dipole moments between the $X^1\Sigma^+$, $A^1\Sigma^+$, $B^1\Pi$, $C^1\Sigma^+$, and $D^1\Sigma^+$ states of the LiH molecule are computed based on the CI wavefunctions obtained, as presented in Figs. 2(a) and 2(b). In Fig. 2(a), the transition dipole moment of $A^1\Sigma^+$–$X^1\Sigma^+$ has a maximum value of $R = 7.20$ a.u., while those for $A^1\Sigma^+$–$X^1\Sigma^+$ and $B^1\Pi$–$X^1\Sigma^+$ converge to constant at large internuclear distances, representing the transition of Li. The absolute values of the transition dipole moments for $C^1\Sigma^+$–$X^1\Sigma^+$ and $D^1\Sigma^+$–$X^1\Sigma^+$ are smaller than those for $A^1\Sigma^+$–$X^1\Sigma^+$. As shown in Fig. 2(b), the transition dipole moments of different excited states are also presented. The transition dipole moment of $D^1\Sigma^+$–$C^1\Sigma^+$ exhibits obvious changes at $R \geq 22.2$ Å, owing to the avoided crossing between the $D^1\Sigma^+$ and $C^1\Sigma^+$ states. The consecutive crossings of the transition dipole moment curves are related to the avoided crossing points at $R = 7.16$ a.u., as verified in previous investigations.[55,56]

The permanent dipole moments for the $X^1\Sigma^+$, $A^1\Sigma^+$, $B^1\Pi$, $C^1\Sigma^+$ and $D^1\Sigma^+$ states are depicted

in Fig. 2(c). Note that many intersections occur for $X^1\Sigma^+$, $A^1\Sigma^+$, $B^1\Pi$, $C^1\Sigma^+$, and $D^1\Sigma^+$ states with an increase in R. The permanent dipole moment of the $X^1\Sigma^+$ state has a small negative value, and approaches to zero with an increase in internuclear distances. The permanent dipole moments of $A^1\Sigma^+$ go from positive to negative at $R = 4.8$ Å, converging to zero at large internuclear distances. With reference to the $B^1\Pi$ state, the values of the permanent dipole moment are small, and remain relatively unchanged regardless of internuclear distance. Regarding the permanent dipole moments of the $C^1\Sigma^+$ and $D^1\Sigma^+$ states, two pairs of significant complementary changes are found around the avoided crossing of $R = 5.0$ Å and $R = 22.2$ Å, respectively, after which the wavefunctions of $C^1\Sigma^+$ and $D^1\Sigma^+$ are exchanged.

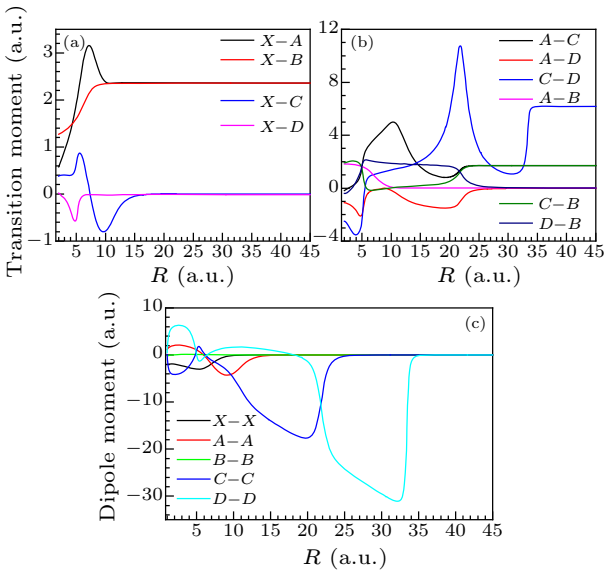


Fig. 2. Transition dipole moments for different states of ^{7}LiH.

Molecular Opacities Calculations. Based on the above calculations of vibration-rotation energy levels, wavefunctions and PECs, the partition functions for the ^{7}LiH molecule, together with the isotopes of ^{7}LiD, ^{6}LiH, and ^{6}LiD are calculated in the temperature range of $T = 10$–2000 K using formula (8) (also see Ref. [57]) via summation over all vibration-rotation levels for the five low-lying bound states. The partition functions of ^{7}LiH and its isotopic molecules are presented in Fig. S1 of the Supplementary Material versus temperature T, and the corresponding partition functions obtained by Diniz *et al.*[37] are also predicted in Fig. S1 of the Supplementary Material. In addition, the partition functions are shown to be in excellent agreement over the temperature range under consideration.

With regard to partition functions, together with other parameters such as transition dipole moments etc., the molecular opacities of ^{7}LiH are computed using Eq. (7). In this work, the line is profiled by means of a Voigt function, comprising the Doppler and collisional broadening effect, considered at a pressure of 10 atm, which is highly relevant in relation to stars and the early Universe,[58,59] and at temperatures of 300 K, 1000 K, 1500 K, and 2000 K. As shown in Figs. 3(a)–3(d), the spectra include five bands resulting from the transitions of $X^1\Sigma^+$–$X^1\Sigma^+$, $X^1\Sigma^+$–$A^1\Sigma^+$, $X^1\Sigma^+$–$B^1\Pi$, $X^1\Sigma^+$–$C^1\Sigma^+$, and $X^1\Sigma^+$–$D^1\Sigma^+$, respectively. It is evident that all spectra are located in the ultraviolet and far-infrared ranges for both ^{7}LiH and ^{7}LiD. In Figs. 3(a) and 3(b), each band of the spectra at 300 K and 1000 K can be clearly identified. The band systems are formed from the transitions of $X^1\Sigma^+$–$X^1\Sigma^+$, $A^1\Sigma^+$–$X^1\Sigma^+$, $B^1\Pi$–$X^1\Sigma^+$, $C^1\Sigma^+$–$X^1\Sigma^+$, and $D^1\Sigma^+$–$X^1\Sigma^+$, from right to left, respectively. For the system of the $X^1\Sigma^+$–$X^1\Sigma^+$ band shown in Fig. 3(a), the vibrational bands for the ^{7}LiH of $\Delta\nu = 1$, 2 and 3 are located at approximately 8.3 μm, 7.4 μm, and 3.7 μm, while the vibrational bands of $\Delta\nu = 1$, 2 and 3 for ^{7}LiD are located at approximately 10.2 μm, 9.4 μm and 4.8 μm, respectively. The pure rotational spectra of $\Delta\nu = 0$ are those for the wavelength $\lambda > 30$ μm for both ^{7}LiH and ^{7}LiD. As shown in Fig. 3(a) for the ^{7}LiH molecule, the peak of the cross section for $X^1\Sigma^+$–$X^1\Sigma^+$ is 4.65×10^{-17} cm^2, with a wavelength of 110 μm. Our calculated result is consistent with the value of 3.34×10^{-17} cm^2 at the wavelength of 100 μm previously reported by Coppola *et al.*,[35] as well as that of the Exomol project,[60] which aims to provide line lists of the important molecular structures[61–63] in cooler stars and extrasolar planets. The difference for the peak of cross section between our work and previously recorded values[35,60] may originate from the accuracy of PECs and TDMs. The peak of opacities originating from the transitions of $A^1\Sigma^+$–$X^1\Sigma^+$, $B^1\Pi$–$X^1\Sigma^+$, $C^1\Sigma^+$–$X^1\Sigma^+$, and $D^1\Sigma^+$–$X^1\Sigma^+$ emerge separately at around 358.3 nm, 279.8 nm, 211.9 nm, 199.8 nm, respectively, for ^{7}LiH, and 345.1 nm, 288.3 nm, 215.1 nm, 201.7 nm, respectively, for ^{7}LiD. Figure 3(a) illustrates that the transitions are weak in the wavelength region of 500–1800 nm, which covers the visible light (VIB) and near infrared (NIR) regions. With an increase in temperature, new features appear in the VIB and NIR regions, while the boundaries between different states in the VIB and NIR regions become increasingly indistinct, due to the increasing population of excited electronic and vibrational states. Figure 4 depicts the calculated opacities of the ^{6}LiH and ^{6}LiD molecules, with their spectra primarily located in the range of the ultraviolet and far-infrared regions at temperatures of 300 K, 1000 K, 1500 K, and 2000 K. The bands of the spectra for ^{6}LiH and ^{6}LiD can be clearly distinguished in Figs. 4(a) and 4(b) at 300 K and 1000 K. As depicted in Fig. 4(a), the pure rotational spectra of $\Delta\nu = 0$ for ^{6}LiH and ^{6}LiD both occur at a wavelength of $\lambda > 20$ μm. The vibrational bands of $\Delta\nu = 1$, 2 and 3 are separately located at approximately 7.9 nm, 7.4 nm, 3.7 nm and 9.9 nm, 9.2 nm, and 4.8 nm. The peak of opacities, generated by the transitions of $A^1\Sigma^+$–$X^1\Sigma^+$, $B^1\Pi$–$X^1\Sigma^+$, $C^1\Sigma^+$–$X^1\Sigma^+$, and

$D^1\Sigma^+$–$X^1\Sigma^+$, appear at around 376.2 nm, 337.1 nm, 224.2 nm, 210.3 nm for ^{6}LiH, and 347.8 nm, 288.7 nm, 211.6 nm, and 199.8 nm for ^{6}LiD, respectively. With increasing temperature, the boundaries, which are distinct at low temperature, become blurred, due to the increasing populations of excited states.

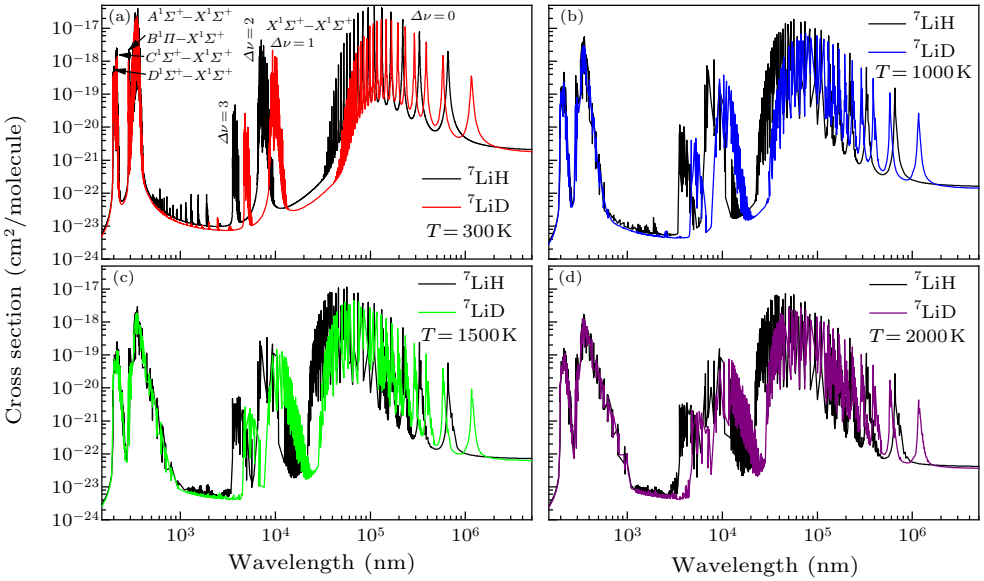


Fig. 3. Opacities of ^{7}LiH and ^{7}LiD ro-vibrational transitions at temperatures of (a) 300 K, (b) 1000 K, (c) 1500 K and (d) 2000 K and at a pressure of 10 atm.

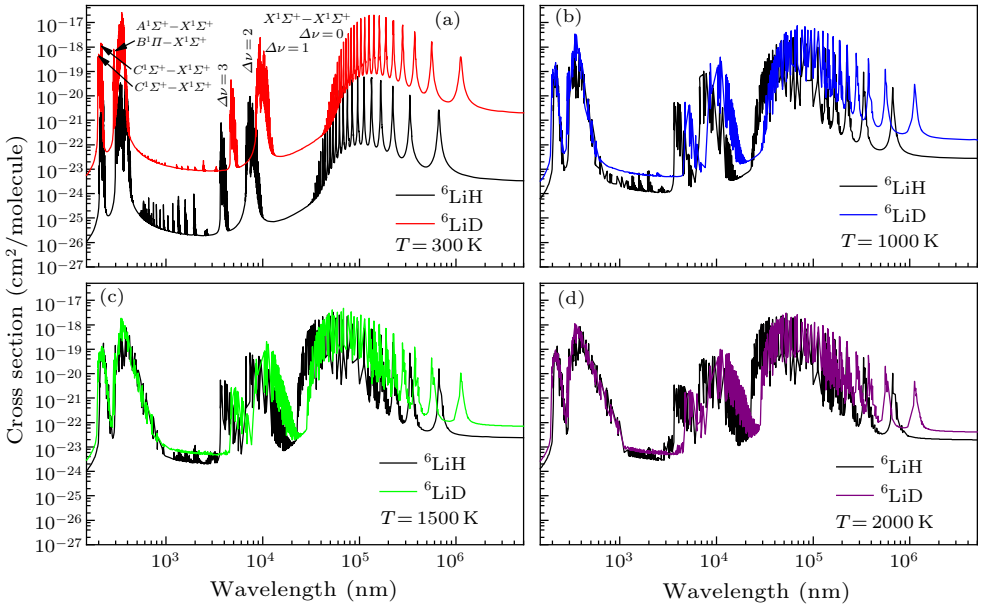


Fig. 4. Opacities of ^{6}LiH and ^{6}LiD ro-vibrational transitions at temperatures of (a) 300 K, (b) 1000 K, (c) 1500 K and (d) 2000 K and at pressure of 10 atm.

In summary, we have calculated ro-vibrational transition lines for the five electronic states of the ^{7}LiH molecule, based on PECs obtained via high-level MRDCI calculations. Using the PECs obtained, spectroscopic constants and vibrational energy levels are computed by solving the Schrödinger equation of diatomic molecules, and good agreements are achieved in comparison with current results available. Partition functions have been calculated for an isotopic molecule of ^{7}LiH, together with its isotopic molecules over a temperature range of 10–2000 K. The opacities of ^{7}LiH and its isotopic molecules have been calculated at temperatures of 300 K, 1000 K, 1500 K, and 2000 K for the conditions of 10 atm. At lower temperatures of 300 K and 1000 K, the electronic transitions of $A^1\Sigma^+$–$X^1\Sigma^+$ and $B^1\Pi$–$X^1\Sigma^+$, and $C^1\Sigma^+$–$X^1\Sigma^+$ and $D^1\Sigma^+$–$X^1\Sigma^+$ can readily be distinguished, and the vibrational transition with $\Delta\nu = 0$, 1, 2 and 3 for $X^1\Sigma^+$–$X^1\Sigma^+$ can also be clearly identified. It is apparent that different band spectra mingle with one another, and that band boundaries become blurred with increasing temperature, due to the rise in populations associated with excited states. Molecular opacity calculations for ^{7}LiH and its isotopic components are performed for the first

five low-lying states, and obvious isotopic effects are identified. This work should provide useful and reliable data for future studies in the fields of astrophysics and fusion science.

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Supplementary Information: Molecular Opacity Calculations for Lithium Hydride at Low Temperature

Gui Ying Liang(梁桂颖)[1], Yi Geng Peng(彭裔耕)[2], Rui Li(李瑞)[3], Yong Wu(吴勇)[1,4**], Jian Guo Wang(王建国)[1]

[1]*Institute of Applied Physics and Computational Mathematics, Beijing 100088, China*

[2] *Department of Applied Physics, Nanjing University of Science and Technology, Nanjing 210094, China*

[3]*Department of Physics, College of Science, Qiqihar University, Qiqihar 161006, China*

[4]*HEDPS, Center for Applied Physics and Technology, Peking University, Beijing 100084, China*

Table S1. The exponents of the diffuse functions used to describe the excited states of lithium and hydrogen atoms

	S	P	D
Li	0.0159	0.0059	0.0091
	0.0053	0.0021	0.0029
	0.0018	0.0011	0.0011
H	0.0148	0.0282	0.0633
	0.0049	0.0094	0.0211
	0.0017	0.0031	0.0071

Table S2. The calculated vibrational level spaceings ΔE_v for the $X^1\Sigma^+$ state of ^{7}LiH (in cm^{-1}).

v	ΔE_v for the $X^1\Sigma^+$ state				
0	Present(cm^{-1})	Expt.[a]	CI[b]	Full CI[c]	Full CI[d]
1	1360.40	1359.71	1359.71	1360.58	1360.3
2	1313.16	1314.89	1314.85	1315.53	1315.26
3	1271.12	1270.89	1270.91	1271.42	1271.17
4	1226.26	1227.77	1227.81	1228.16	1227.93
5	1184.04	1185.44	1185.45	1185.66	1185.45
6	1145.82	1143.77	1143.75	1143.84	1143.64
7	1103.62	1102.17	1102.57	1102.54	1102.38
8	1062.45	1061.78	1061.77	1061.65	1061.5
9	1022.18	1021.18	1021.18	1020.97	1020.85
10	981.22	980.52	980.58	980.29	980.2
11	940.24	939.62	939.69	939.36	939.3
12	899.53	898.09	898.16	897.79	897.77
13	856.16	855.48	855.52	855.17	855.18
14	812.57	811.19	811.18	810.87	810.94
15	763.63	764.44	764.40	764.17	764.27
16	716.53	714.12	714.21	714.03	714.18
17	661.53	659.37	659.31	659.18	659.36
18	598.39	597.89	598.01	597.92	598.13
19	529.50	527.91	527.98	528.07	528.31
20	447.47	446.61	446.23	446.99	447.23
21	351.44	350.99	349.83	351.88	352.12
22	238.08	237.71	239.56	240.14	240.41

[a][1]

[b][2]

[c][3]

[d][4]

Table S3. The calculated vibrational level spaceings ΔE_v for the $A^1\Sigma^+$ state of ^{7}LiH (in cm^{-1}).

v	ΔE_v for the $A^1\Sigma^+$ state				
0	Present	Expt.[a]	Full CI[b]	TDCI[c]	SCF+CI[d]
1	283.69	280.84	278.43	274.91	295.5
2	316.45	312.69	310.93	309.47	326.0
3	334.34	335.69	333.96	333.52	347.6
4	350.96	352.79	351.32	350.80	362.7
5	364.51	365.85	364.47	363.29	374.8
6	374.82	375.60	374.40	372.55	384.7
7	380.32	382.68	381.70	379.38	391.9
8	384.71	387.55	386.58	383.99	396.3
9	390.12	390.37	389.59	386.73	398.8
10	389.79	391.05	390.79	388.04	399.8
11	389.94	389.19	390.37	387.76	399.2
12	388.26	385.94	388.55	385.99	397.0
13	385.21	381.32	385.33	382.86	393.5
14	380.65	375.21	380.82	378.17	389.0
15	375.45		375.02	371.86	383.1
16	369.48		367.88	363.81	375.8
17	361.56		359.31	353.70	367.1
18	351.58		349.11	341.17	356.8
19	340.70		337.00	325.67	344.8
20	326.92		322.47		330.4
21	310.50		304.78		313.1
22	290.15		282.75		291.8
23	264.26				264.4
24	230.08				228.3
25	184.03				

[a][5]

[b][3]

[c][6]

[d][7]

Table S4. The calculated vibrational level spaceings ΔE_v for the $B^1\Pi$ state of ^7LiH (in cm^{-1}).

ν	ΔE_v for the $B^1\Pi$ state			
0	Present	Expt.[a]	TDCI[b]	SCF+CI[c]
1	131.32	130.73	66.45	133.4
2	33.50	45.9		34.7

[a][8]

[b][6]

[c][7]

Table S5. The calculated vibrational level spaceings ΔE_v for the $C^1\Sigma^+$ state of ^{7}LiH (in cm^{-1}).

v	ΔE_v for the $C^1\Sigma^+$state			
0	Present	Expt.[a]	Full+CI[b]	CI[c]
1	289.08	290.99	290.98	285.04
2	285.95	286.42	286.56	284.38
3	280.90	281.79	281.93	279.83
4	276.40	277.13	277.22	273.71
5	271.40	272.44	272.57	267.68
6	266.36	267.73	267.92	262.43
7	261.85	269.03	263.25	258.04
8	257.38	258.31	258.50	253.92
9	252.58	253.63	253.75	249.99
10	247.83	248.19	249.08	246.12
11	243.33	244.33	244.45	242.30
12	238.89	23.729	239.84	238.50
13	234.27	235.17	235.26	234.67
14	229.73	230.67	230.74	230.80
15	225.42	226.21	226.27	226.85
16	221.12	221.81	221.85	222.85
17	216.78	217.48	217.48	218.82
18	212.51	213.19	213.18	214.82
19	208.39	205.98	208.96	210.82
20	204.29	204.81	204.80	206.82
21	200.22	200.710	200.71	202.82
22	196.22	196.67	196.66	198.82
23	192.35	192.68	192.67	194.81
24	188.49	188.73	188.72	190.77
25	184.61	184.84	184.81	186.70

26	180.76	180.97	180.92	182.56
27	177.02	177.16	177.07	178.35
28	173.25	173.35	173.23	174.Q2
29	169.53	169.57	169.39	169.55
30	166.09	165.80	165.50	164.87
31	163.51		161.48	159.91

[a][9] and [10]

[b][3]

[c][11]

Table S6. The calculated vibrational level spaceings ΔE_v for the $D^1\Sigma^+$ state of ^{7}LiH (in cm^{-1}).

v	ΔE_v for the $D^1\Sigma^+$state		
0	Present	Expt.[a]	Full CI[b]
1	59.01	59.27	58.07
2	57.08	56.81	58.35
3	58.11	59.74	57.83
4	56.97	57.19	56.64
5	44.21		42.94
6	19.01	11.20	11.68
7	55.40	50.31	51.39
8	41		44.24
9	22.58		25.58
10	33.01		32.36

[a][12]

[b][3]

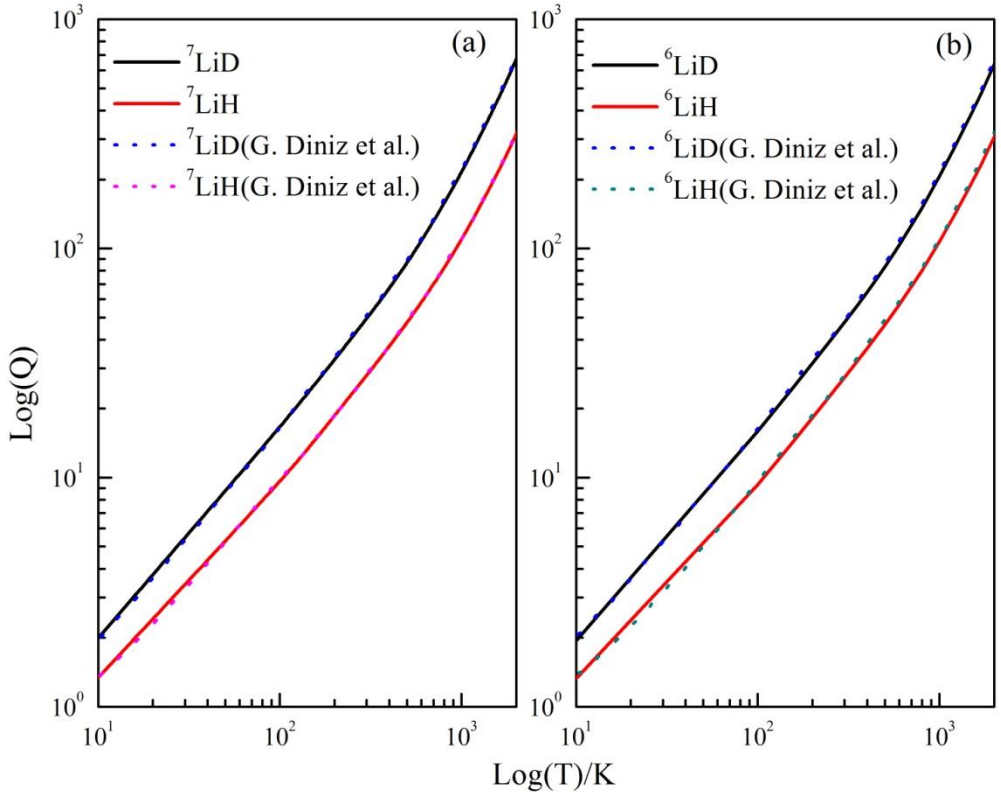


Figure S1. The partition function for ^{7}LiH, ^{7}LiD, ^{6}LiH and ^{6}LiD.

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